

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Solve an ODE Numerically

1. Euler’s Method

Euler’s Method is a numerical method for approximating the solution of a first-order initial value problem of the form:

$$y' = f(t, y), \quad y(t_0) = y_0.$$

Instead of solving this analytically (using calculus), we use small steps to estimate values of y . If we choose a *step size* h , then starting from (t_0, y_0) , we compute:

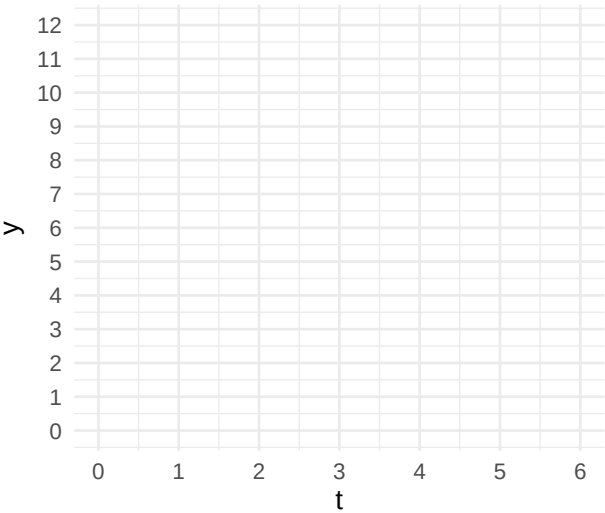
$$y_{n+1} = y_n + hf(t_n, y_n) \quad \text{and} \quad t_{n+1} = t_n + h.$$

This builds a table of points $(t_0, y_0), (t_1, y_1), \dots$ that approximates the solution curve.

Consider the autonomous ODE $y' = y \left(1 - \frac{y}{10}\right)$ with initial condition $y(0) = 1$.

- a. Let $f(t_n, y_n) = y_n \left(1 - \frac{y_n}{10}\right)$. The table below shows a partially worked example using Euler’s method to approximate the solution of the ODE using $h = 2$. Complete the table.
- b. Use the blank axis shown below to plot the points (t_n, y_n) and connect them.

n	t_n	y_n	$f(t_n, y_n)$	$y_{n+1} = y_n + hf$
0	0	1	0.9	$1 + 2(0.9) = 2.8$
1	2	2.8	2.016	$2.8+2(2.016)=6.832$
2	4	6.832	2.1644	
3	6			



2. Approximating Solutions

Continued from Problem (1). Let $f(t_n, y_n) = y_n \left(1 - \frac{y_n}{10}\right)$.

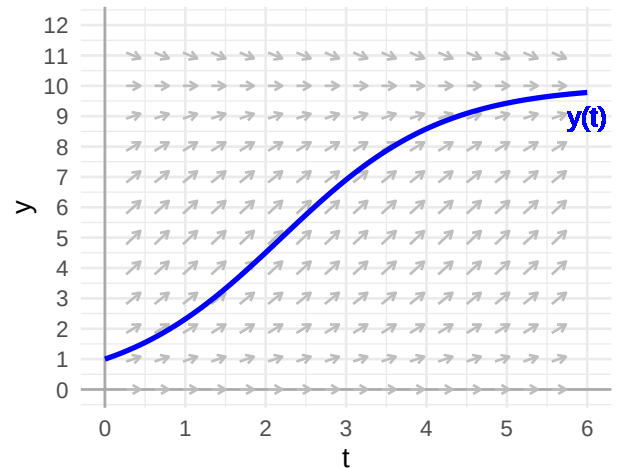
- a. The table below shows a partially worked example using Euler's method to approximate the solution of the ODE using $h = 1$. Complete the table.

n	t_n	y_n	$f(t_n, y_n)$	$y_{n+1} = y_n + hf$
0	0	1	0.9	$1 + 1(0.9) = 1.9$
1	1	1.9	1.539	$1.9 + 1(1.539) = 3.439$
2	2	3.439	2.2563	
3	3			
4	4			
5	5			
6	6			

- b. The table below shows a partially worked example using Euler's method to approximate the solution of the ODE using $h = 0.5$. Complete the table.

n	t_n	y_n	$f(t_n, y_n)$	$y_{n+1} = y_n + hf$
0	0	1	0.9	$1 + 0.5(0.9) = 1.45$
1	0.5	1.45	1.2398	$1.45 + 0.5(1.2398) = 2.0699$
2	1	2.0699	1.6415	
3	1.5			
4	2			
5				
6				
7				
8				
9				
10				
11				
12				

- c. On the slope field below, it shows a specific solution of the given ODE. Stitch together several segments to produce a graph of approximated solutions for each table completed in Parts (a-b) and Problem (1b), label each graph accordingly.



- d. In Part (c), which graph is more accurate and why? What happens to the graph as the step size gets smaller and smaller?